SETUP | BEFORE CLASS

Prepare R once; test the full path

Extract the **complete** student ZIP into a writable folder. Keep its directory structure. Open `Pascall_student.Rproj` in RStudio. RStudio is convenient, not compulsory.

Installation

```
source("00_install_check.R")
source("RUN_R_CHECKS.R")
```

The first script checks input hashes and installs missing **ape**, **phytools** and **treedater** packages into the local `.R-library` folder. Installation requires an internet connection; the practical itself uses local files.

Use an up-to-date R installation. Windows/macOS binary packages are usually simplest; if R reports a source-compilation requirement, use the appropriate R build tools or the explicit rescue instead of debugging during class. No PhyML executable, GenBank access or sequence aligner is needed for the core practical.

What counts as ready?

The second script runs all five live blocks and checks IDs, distances, dates, rates, monophyly and output creation. It prints **READY** only after success and writes `validation/NATIVE_READY.txt`. Package installation alone is not readiness.

> Release validation: the native R route, both rescue routes and optional extensions passed on Linux. Run RUN_R_CHECKS.R before class. READY checks execution and numerical consistency, not the biological adequacy of a clock model.

A successful check creates a validated native snapshot in rescue/native/. Before a new check, the previous cache is archived under rescue/previous_* and the old READY status is removed. Failure leaves no active cache; it never silently selects a reference result.

Without RStudio

```
# From the extracted project folder, in a terminal:
Rscript 00_install_check.R
Rscript RUN_R_CHECKS.R
Rscript run_without_rstudio.R
```

Do not run the scripts from inside an unopened ZIP, a read-only directory or a different working directory.

Three clearly labelled routes

Route	What it actually does
Normal R	Runs rooting, OLS diagnostics, strict/relaxed dating and branch-rate extraction. Requires all three packages.
Native R rescue	Reads the validated native snapshot included in this release. Inputs, code and numerical contents are checked before replay.
Reference/ browser	Uses independently Python-rooted reference trees and published figures. No native dating is claimed.

Explicit rescue in R

Native rescue is the default explicit R rescue and requires the included validated cache. To use the independent reference route, set options(pascall.rescue_mode="reference") before sourcing Pascall_rescue.R. There is no automatic fallback.

```
# Install ape alone if the normal package installation is blocked:
install.packages("ape", repos="https://cloud.r-project.org")
source("Pascall_rescue.R")
```

No R available?

Open `docs/Pascall_offline_EN.html` in a browser. It is self-contained and works without internet. It provides all five stages, visible R code, reference figures, locally saved notes and a notes download. The temporal tab recalculates descriptive OLS in JavaScript when the selected reference root changes.

> The browser does not infer roots or fit molecular clocks. It clearly identifies the precomputed Python root and the published BEAST figure. Switching a selector must not be mistaken for executing an R inference.

Find your results

Live outputs go to a new `runs/live_…/` folder: PDF/PNG figures, CSV tables, the complete `lesson.rds`, dated Newick and session information. `runs/latest_run.txt` identifies the latest completed live run.

On failure, read `validation/native_run.log` and `validation/native_error.txt`. Do not silently replace a failed analysis with cached results. A later successful run removes a stale error report.